

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: df=pd.read_csv('diabetes.csv')
```

```
In [3]: df
```

```
Out[3]:
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction
0	6	148	72	35	0	33.6	0.627
1	1	85	66	29	0	26.6	0.351
2	8	183	64	0	0	23.3	0.672
3	1	89	66	23	94	28.1	0.167
4	0	137	40	35	168	43.1	2.288
...	...	...	...	...	...	...	...
763	10	101	76	48	180	32.9	0.171
764	2	122	70	27	0	36.8	0.340
765	5	121	72	23	112	26.2	0.245
766	1	126	60	0	0	30.1	0.349
767	1	93	70	31	0	30.4	0.315

768 rows × 9 columns

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Pregnancies         768 non-null   int64
1   Glucose             768 non-null   int64
2   BloodPressure       768 non-null   int64
3   SkinThickness       768 non-null   int64
4   Insulin             768 non-null   int64
5   BMI                 768 non-null   float64
6   DiabetesPedigreeFunction 768 non-null   float64
7   Age                 768 non-null   int64
8   Outcome             768 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: Pregnancies      0
        Glucose          0
        BloodPressure    0
        SkinThickness     0
        Insulin           0
        BMI               0
        DiabetesPedigreeFunction  0
        Age              0
        Outcome           0
        dtype: int64
```

```
In [7]: X = df.drop('Outcome', axis=1)
        y = df['Outcome']
```

```
In [8]: from sklearn.model_selection import train_test_split

        X_train, X_test, y_train, y_test = train_test_split(
            X, y, test_size=0.2, random_state=42
        )
```

```
In [9]: from sklearn.preprocessing import StandardScaler
        sc = StandardScaler()
        X_train = sc.fit_transform(X_train)
        X_test = sc.transform(X_test)
```

```
In [10]: from sklearn.naive_bayes import GaussianNB
         gaussian = GaussianNB()
         gaussian.fit(X_train, y_train)
```

```
Out[10]: ▾ GaussianNB
         GaussianNB()
```

```
In [11]: Y_pred = gaussian.predict(X_test)
```

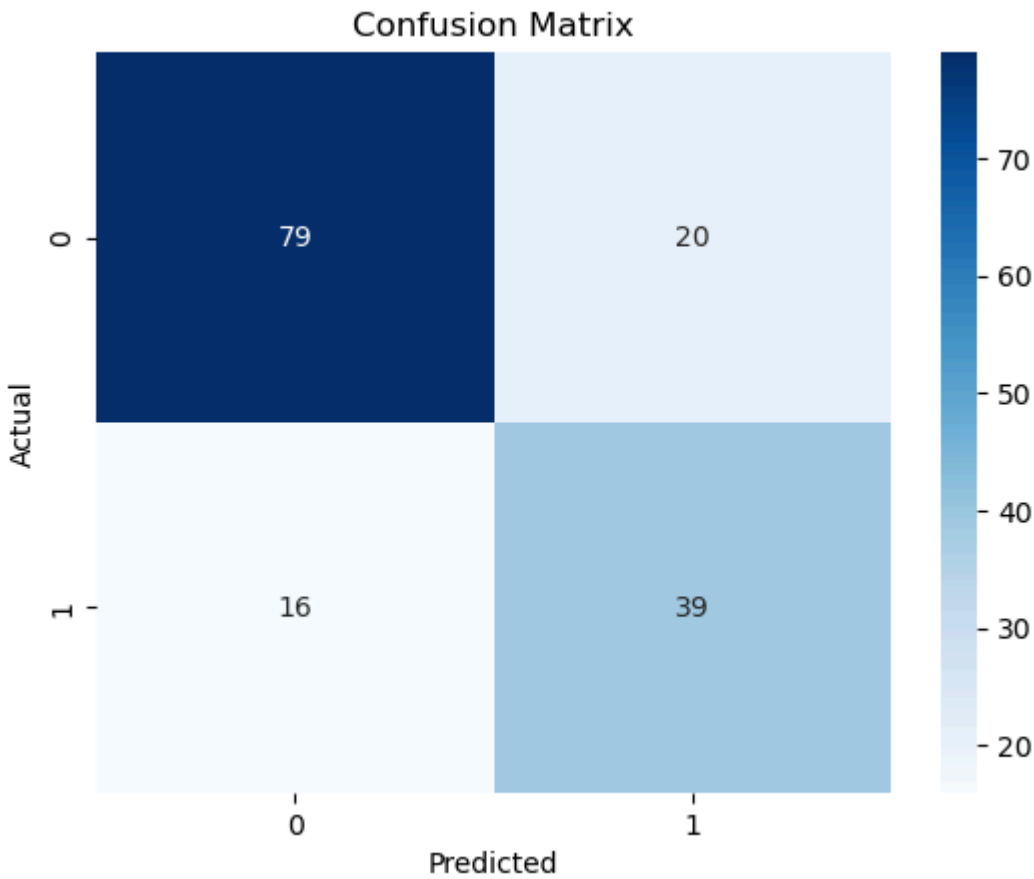
```
In [12]: from sklearn.metrics import accuracy_score, precision_score, recall_score, confusion_matrix

         accuracy = accuracy_score(y_test, Y_pred)
         precision = precision_score(y_test, Y_pred)
         recall = recall_score(y_test, Y_pred)
         cm = confusion_matrix(y_test, Y_pred)

         print("Accuracy:", accuracy)
         print("Precision:", precision)
         print("Recall:", recall)
         print("Confusion Matrix:\n", cm)
```

```
Accuracy: 0.7662337662337663
Precision: 0.6610169491525424
Recall: 0.7090909090909091
Confusion Matrix:
[[79 20]
 [16 39]]
```

```
In [13]: import seaborn as sns
         sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
         plt.xlabel("Predicted")
         plt.ylabel("Actual")
         plt.title("Confusion Matrix")
         plt.show()
```



```
In [ ]:
```